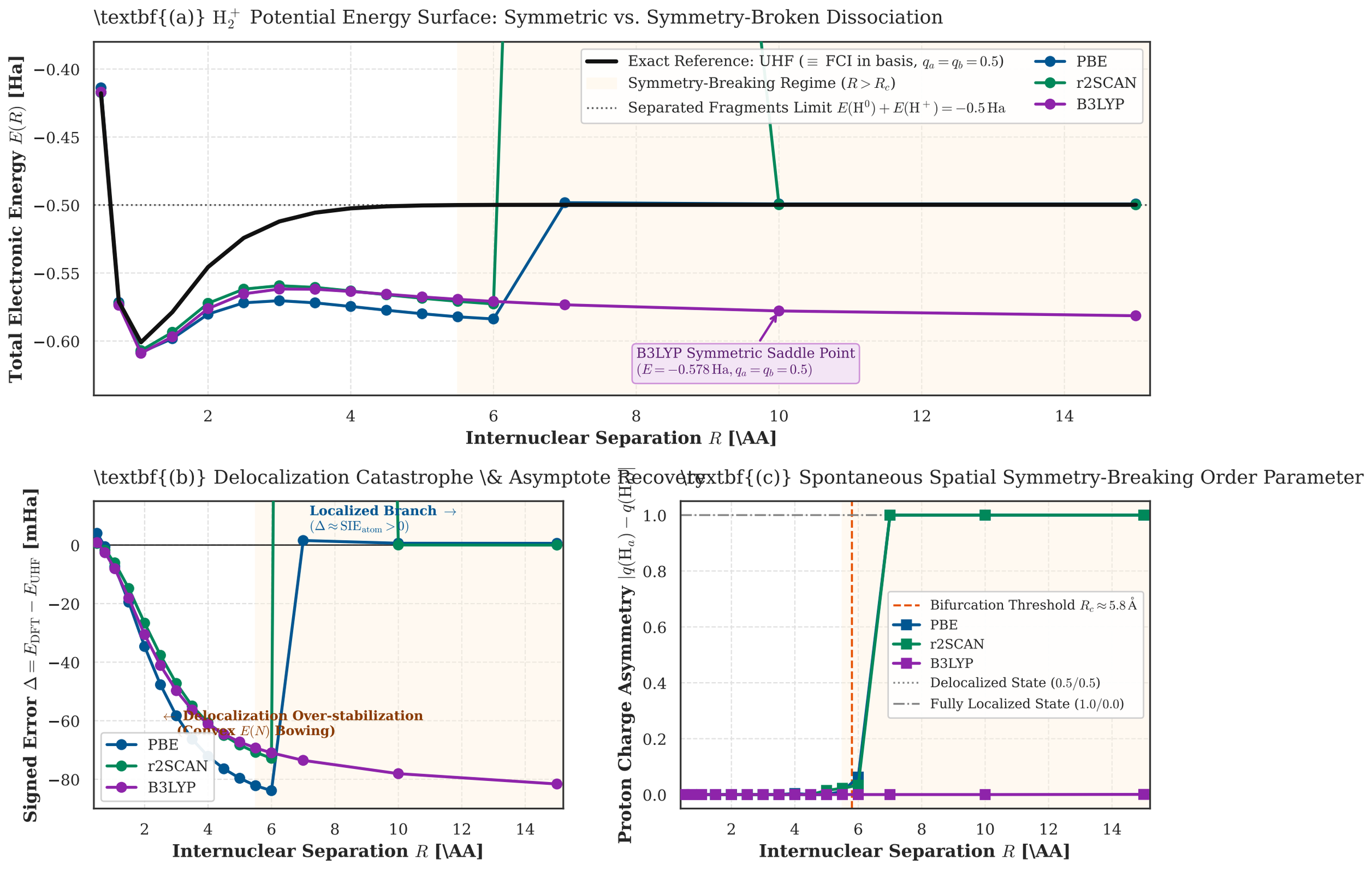

\textbf{(a)} H_2^+ Potential Energy Surface: Symmetric vs. Symmetry-Broken Dissociation
Total Electronic Energy $E(R)$ [Ha]
−0.40
−0.45
−0.50
−0.55
−0.60
Exact Reference: UHF (≡ FCI in basis, $q_a = q_b = 0.5$)
Symmetry-Breaking Regime ($R > R_c$)
Separated Fragments Limit $E(H^0) + E(H^+) = -0.5$ Ha
PBE
r2SCAN
B3LYP
B3LYP Symmetric Saddle Point
($E = -0.578$ Ha, $q_a = q_b = 0.5$)
2
4
6
8
10
12
14
Internuclear Separation R [\AA]
\textbf{(b)} Delocalization Catastrophe \& Asymptote Recovery
Signed Error $\Delta = E_{DFT} - E_{UHF}$ [mHa]
0
−20
−40
−60
−80
Localized Branch →
($\Delta \approx$ SIE$_{atom} > 0$)
← Delocalization Over-stabilization
(Convex $E(N)$ Bowing)
PBE
r2SCAN
B3LYP
2
4
6
8
10
12
14
Internuclear Separation R [\AA]
\textbf{(c)} Spontaneous Spatial Symmetry-Breaking Order Parameter
Proton Charge Asymmetry $|q(H_a) - q(H_b)|$
1.0
0.8
0.6
0.4
0.2
0.0
Bifurcation Threshold $R_c \approx 5.8$ Å
PBE
r2SCAN
B3LYP
Delocalized State (0.5/0.5)
Fully Localized State (1.0/0.0)
2
4
6
8
10
12
14
Internuclear Separation R [\AA]